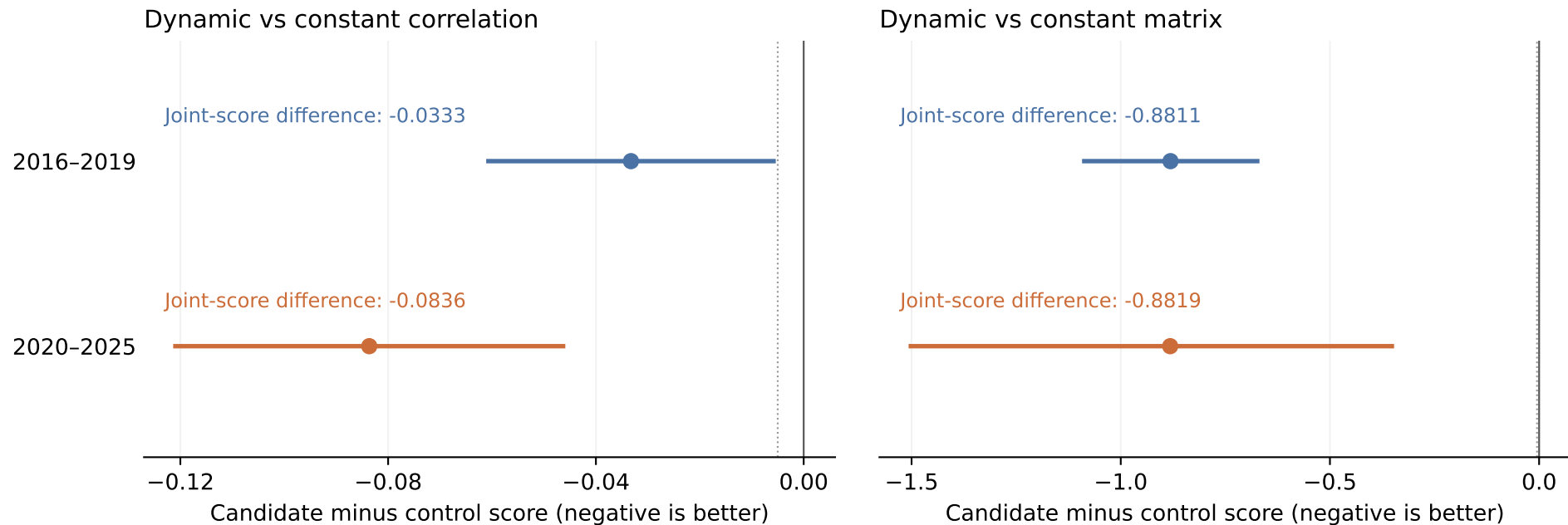


Does changing dependence improve joint QQQ-SPX risk forecasts?



Lines: nominal 95% block-bootstrap/HAC interval envelopes. Unhalved matrix score; no percentage gains.

Both controls require ≥ 0.005 improvement in both periods, stability and multiplicity gates. Archival ETF/index daily returns.